

Disposable Expiration Valves

Offering a hygienic and cost effective solution

In 2009 London's St Mary's Hospital Imperial NHS Trust changed from reusable to disposable expiration valves in the adult ICU. The decision, which was based on several factors including improved infection control and ease of use, has also brought the benefits of lower cost and immediate availability.



When St Mary's Hospital built a new ICU unit and looked into new equipment in 2007, Dräger won the tender to supply new Evita XL ventilators. The move to 100% disposable valves was made at the same time in light of the new sterilisation process. Expiration valves safeguard the functions of Evita ventilators and control the breathing phases.

Mr Brian Dornan, Senior Critical Care Scientist, Imperial NHS Trust, St Mary's Hospital London, explains why the hospital changed to Dräger disposable expiration valves and the improvements they have found in both staff as well as patient welfare.

*"Other advantages lie in both **cost and time savings**. For instance, when we used the reusable version we had to autoclave each valve after use. Originally this was carried out on site but, with the announcement of the national directive to go to Regional Central Sterile Supply Departments (RCSSD), we found that often the valve would not return for 24 hours. This meant that, instead of having a stock of three valves per ventilator, we had to allow for a turnaround of perhaps six."*

"In addition to the time and money spent on this sterilisation, which could vary wildly from £60 to £0.60p (approx. €70 to €0.70) depending on the numbers involved in the individual sterilisation cycle, the valves would occasionally come back damaged. This meant that someone had to check them prior to use to ensure that they had been properly assembled. Parts could be lost and, unlike the disposable version, the reusables tended to discolour and wear with age."

*"With the disposable expiration valve, the valve is **ready for use** and the single use cost is very easy to calculate. What's more, their comparatively low cost means that an adequate quantity can be held in stock, ready for immediate use."*

St Mary's Hospital had used Dräger Evita 2 dura ventilators in the past and the disposable valves had been tested on those with very positive results.

He adds:

"We treat between 500 and 600 adult patients in ICU each year. As a minimum, we change the valve after each patient. Where patients are on continuous nebulisation with Prostacycline or NAC (N-Acetylcysteine), which can cause problems with crystallisation on the expiration valve's membrane, we simply change the valve more frequently."

*"Taken product performance of disposable Evita expiration valves, we have found that there is no difference at all between the ventilation quality of the reusable and disposable versions, the **quality is identical between both expiration valves**."*



Evita XL ventilator in the ICU



Changing the Disposable Expiration Valve



Disposable Expiration Valve for Evita ventilators

He continues:

*“There are a number of advantages associated with disposability, particularly where the **prevention of cross contamination** is concerned. In terms of the patient benefit, the valve has not been used by another patient and the **risk of infection transfer is minimised**. For the nursing staff, the daily routine is made safer and easier because the disposable expiration valve requires no assembly.*

*“Ensuring that infection control and hygiene standards are met, the valve is **easily and quickly replaced** for each patient and then is just as easily disposed of. As far as I am concerned, by using the Dräger disposable expiration valve I know that I have gone as far as I can to prevent an infection.”*

With the number of nosocomial infections on the increase, the use of disposable valves is breaking new ground in reducing labour intensive reprocessing costs, improving infection control and saving time whilst maintaining the very best hygienic conditions.

Mr Dornan concludes:

“The average cost of stay per patient on the ICU is £2,500 (approx. €2,900) per day. Compared to the quantity of cost of care the cost per single disposable Evita expiration valve is not a lot. It is a cost, but again it is worth the safety.

*“Our statistics for ventilator associated pneumonia (VAP) at St Mary's is well below the average national UK value and I certainly feel happier in the knowledge that **patients and staff are properly protected**.”*



St Mary's Hospital

HISTORY

St Mary's Hospital is a hospital located in Paddington, London, England that was founded in 1845. This hospital first opened its doors to patients in 1851, the last of the great voluntary hospitals to be founded.

With the shift towards community healthcare delivered in the early 20th century, partly due to the social

medicine revolution, pressure on bed occupancy relaxed, and with the formation of the NHS in the 1940's, many of the local hospitals of the St Mary's teaching hospital group eventually closed and relocated services to the Paddington basin site.

Since the UK's first academic health science centre was created in 2008, it is operated by Imperial College Healthcare NHS Trust.

CRITICAL CARE AT ST MARY'S HOSPITAL

The adult intensive care unit (AICU) with 16 Dräger Evita XL ventilators at St Mary's provides specialist care for critically ill adult patients.

The intensive care team comprises intensive care consultants, medical trainees, intensive care nurses, physiotherapists, a pharmacist and a dietician.

The AICU consultants provide continuous dedicated cover to the unit with all requests for admission referred to the AICU consultant on duty.



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